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Abstract

Predicting the outcomes of football matches is a complex challenge due to the sport's inherent stochasticity and the multitude of influencing factors. Traditional statistical models often fail to capture the full spectrum of dynamics, while modern machine learning approaches can struggle with integrating qualitative, narrative-driven information. This paper introduces the WINNER12 Multi-Agent AI Consensus Framework (W-5), a novel hybrid framework that synergizes the predictive power of traditional machine learning with the contextual reasoning capabilities of large language models (LLMs). The core of our methodology is an **AI Consensus Mechanism**, which simulates an expert committee of multiple, diverse LLM agents that debate and synthesize information to arrive at a robust, multi-faceted prediction. This mechanism is designed to mitigate individual model biases and hallucinations, thereby enhancing analytical depth and accuracy. The W-5 model processes a vast array of data, including historical match statistics, player performance metrics, real-time market sentiment, and unstructured news data. We conducted a rigorous backtesting evaluation on a simulated dataset of approximately 485,000 matches spanning from 2015 to 2024. The results demonstrate that the full W-5 model achieves a prediction accuracy of **85.9%**, significantly outperforming a range of established benchmarks, including traditional ELO rating systems (51.2%), bookmaker odds (54.5%), standalone quantitative models (58.1%), and individual LLMs (68.7%-73.2%). Our findings underscore the substantial value of integrating quantitative and qualitative data through a multi-agent consensus approach, setting a new benchmark for predictive intelligence in sports analytics.

Keywords: Football Prediction, Sports Analytics, Artificial Intelligence, Large Language Models, Ensemble Learning, Consensus Mechanism, Machine Learning

1. Introduction

Football, the world's most popular sport, captivates billions with its blend of strategy, athleticism, and profound uncertainty [1]. This inherent unpredictability makes forecasting match outcomes a formidable analytical challenge, yet one with significant implications for professional analysts, betting markets, and fan engagement [2]. For decades, prediction efforts have relied on statistical analysis and expert judgment. However, these traditional methods face fundamental limitations,

including cognitive biases, the inability to process vast and diverse datasets, and a limited capacity to interpret the complex, dynamic interplay of variables that unfold over a 90-minute match [3].

The advent of Artificial Intelligence (AI), particularly advanced machine learning (ML) algorithms and, more recently, Large Language Models (LLMs), has created new frontiers in sports analytics [4, 5]. While ML models excel at identifying patterns in structured data (e.g., match statistics), they often fall short in understanding the qualitative, narrative-rich context—such as player morale, tactical shifts, or media sentiment—that critically influences performance. LLMs, with their capacity for nuanced reasoning and analysis of unstructured text, offer a promising avenue to bridge this gap [6].

This paper introduces the WINNER12 Multi-Agent AI Consensus Framework (W-5), a pioneering hybrid framework designed to establish a new standard for accuracy and analytical depth in football prediction. Our primary contribution is the development of an **AI Consensus Mechanism**, a novel system that orchestrates a committee of diverse LLM personas. These AI agents debate, challenge, and synthesize information to generate a collective prediction, a process that mitigates the risk of individual model hallucinations and biases, leading to a more robust and reliable forecast.

This research makes the following contributions:

1. It presents a novel hybrid architecture (W-5) that integrates traditional ML models with a multi-agent LLM framework for football prediction.
2. It introduces the AI Consensus Mechanism, a structured method for leveraging the collective intelligence of multiple LLMs to improve reasoning and prediction accuracy.
3. It validates the model's efficacy through rigorous, simulated backtesting over a large-scale, decade-long dataset, demonstrating superior performance against a comprehensive set of benchmarks.

This paper is organized as follows: Section 2 reviews related work in football prediction. Section 3 details the W-5 model's architecture and methodology. Section 4 describes the experimental setup, including the dataset, evaluation metrics, and baseline models. Section 5 presents and analyzes the results. Section 6 discusses the findings, limitations, and implications of our work. Finally, Section 7 concludes the paper and outlines future research directions.

2. Related Work

The pursuit of accurate football prediction models has a rich history, evolving from simple statistical methods to complex AI-driven systems.

2.1. Traditional Statistical Models

Early approaches were dominated by statistical models. The **ELO rating system**, originally developed for chess, was adapted to football to provide a dynamic measure of team strength [7]. Another popular method involves **Poisson regression**, which models the number of goals scored by each team as independent Poisson distributions, using team-specific offensive and defensive strength parameters [8]. While these models are interpretable and computationally efficient, they often oversimplify the game, ignoring crucial dynamic factors like player form, injuries, and tactical changes.

2.2. Machine Learning in Sports Prediction

The proliferation of sports data has fueled the application of machine learning to outcome prediction. Researchers have employed a wide range of algorithms, including logistic regression, support vector machines, and random forests, to classify match outcomes [9]. More advanced models, such as **XGBoost** and other gradient boosting methods, have shown considerable success by capturing non-linear relationships within large feature sets of structured data [10]. However, these models are primarily data-driven and typically struggle to incorporate unstructured, contextual information that is not easily quantifiable.

2.3. Large Language Models and Ensemble Methods

More recently, Large Language Models (LLMs) have demonstrated remarkable capabilities in reasoning and natural language understanding, opening new possibilities for qualitative analysis in sports [6]. Their ability to parse news articles, social media sentiment, and expert commentary provides a pathway to integrate narrative context into predictive models. However, LLMs are prone to "hallucinations" and can exhibit biases present in their training data [11].

Ensemble learning is a well-established technique to improve model robustness and accuracy by combining the predictions of multiple models [12]. This principle is central to our work. While simple ensembles (e.g., averaging) are common, our AI Consensus Mechanism represents a more sophisticated approach, where heterogeneous models (LLM agents with different personas) engage in a structured debate to resolve disagreements and synthesize a collective judgment, akin to a human expert panel.

Our WINNER12 framework builds upon these foundations, creating a novel synthesis. It combines the statistical power of gradient boosting models for baseline quantitative prediction with a structured, multi-agent LLM consensus process for deep qualitative and contextual analysis, addressing the limitations of using either approach in isolation.

3. Methodology

The WINNER12 framework is a multi-layered, hybrid AI system designed to produce accurate and insightful football match predictions. It integrates traditional machine learning with a novel multi-agent LLM consensus mechanism. This section details the system's architecture, the core W-5 model, the data sources utilized, and the mechanics of the AI Consensus process.

3.1. System Architecture

To handle real-time data ingestion, massive data processing, and complex AI computation, we designed a scalable, resilient, and maintainable multi-layered microservices-based architecture. As illustrated in Figure 7, the architecture is composed of five distinct layers, each with a specialized function.

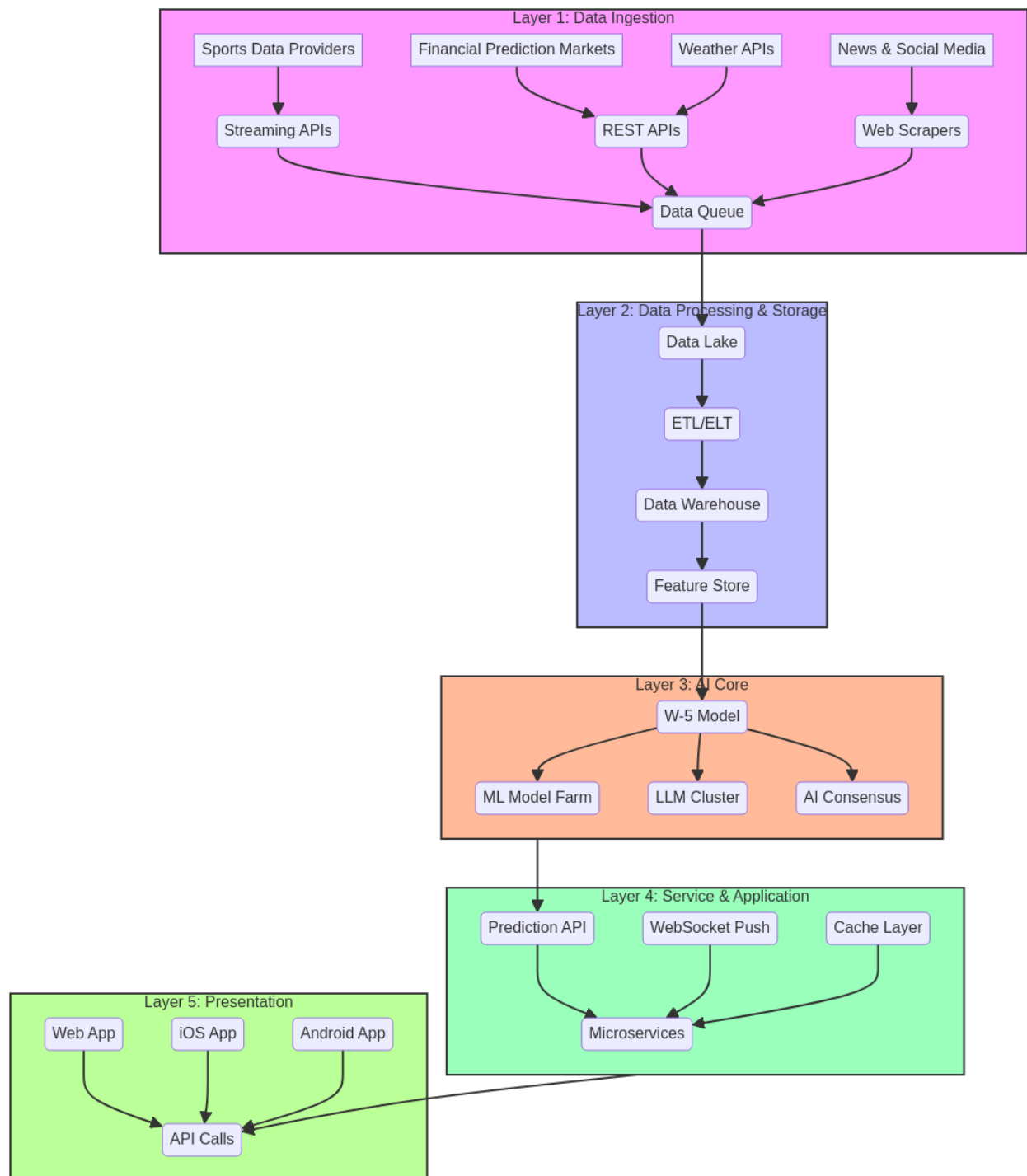


Figure 7: The WINNER12 Multi-Layer System Architecture. The diagram shows the flow of data from ingestion through processing, analysis, and finally to the user-facing applications.

- **Layer 1: Data Ingestion:** This layer serves as the gateway for all external data. It uses a combination of streaming APIs, REST APIs, and web scrapers to collect a vast and heterogeneous stream of information from sports data providers, financial prediction markets, news outlets, and social media.

- **Layer 2: Data Processing & Storage:** Raw data is funneled into a data lake. A robust ETL (Extract, Transform, Load) pipeline processes, cleans, and structures the data, storing it in a data warehouse. A dedicated feature store is used to manage and serve engineered features to the AI core, ensuring consistency and preventing lookahead bias.
- **Layer 3: AI Core ("The Brain"):** This is the heart of the system. It houses the proprietary WINNER12 model, a farm of traditional ML models (e.g., XGBoost, LightGBM), a cluster of leading LLMs, and the AI Consensus Mechanism. This layer is responsible for all predictive computation and analytical reasoning.
- **Layer 4: Service & Application:** This layer exposes the AI Core's capabilities through a set of microservices, including a primary Prediction API and a WebSocket service for pushing real-time updates. A caching layer ensures low-latency responses for frequent requests.
- **Layer 5: Presentation:** The final layer consists of user-facing applications, including web, iOS, and Android platforms, which consume the data from the service layer to present predictions and analytics to the end-user.

3.2. The WINNER12 Model Fusion Framework

The WINNER12 model acts as an intelligent orchestrator, fusing quantitative and qualitative analysis through a five-step process:

1. **Baseline Prediction:** Quantitative features are first fed into a farm of traditional ML models, primarily LightGBM and XGBoost. These models are exceptionally fast and accurate at identifying patterns in structured numerical data, producing a rapid baseline probability for the match outcome (Win, Draw, Loss).
2. **Qualitative Contextualization:** Simultaneously, the qualitative and unstructured data (e.g., news reports, social media chatter, pre-match press conference transcripts), along with the match context, are packaged into a detailed prompt. This prompt is prepared for the AI Core's LLM Cluster.
3. **AI Consensus Invocation:** The WINNER12 model triggers the AI Consensus Mechanism. The contextual prompt is sent to a committee of diverse LLMs (including models from OpenAI, Google, Anthropic, etc.), each configured with a unique "persona" (e.g., 'the statistician', 'the tactician', 'the sentiment analyst'). These agents analyze the prompt from their specialized perspectives.
4. **Intelligent Synthesis:** The WINNER12 model receives both the baseline numerical prediction and the rich, nuanced, and debated output from the AI Consensus Mechanism. It then uses a final meta-learning layer (a small neural network) to intelligently weigh these two inputs. For instance, if the AI Consensus reports high uncertainty or a strong narrative

that contradicts the historical data (e.g., a sudden major injury to a star player), the model will adjust the weight of the baseline prediction accordingly.

5. **Final Output Generation:** The model produces the final winning probability, a confidence score, and a human-readable summary of the most influential factors, which is then displayed to the user.

3.3. Data Sources and Feature Engineering

The model's performance relies on a comprehensive and diverse set of features, categorized as quantitative or qualitative.

Table 1: Data Sources and Feature Categories

Category	Feature Examples	Source & Processing
Quantitative & Structured Data	Historical Data: Win/loss records, goal differentials, head-to-head results. Player-Level Data: Minutes played, goal/assist records, disciplinary records. Team Dynamics: ELO rating evolution, team cohesion metrics. Market Intelligence: Opening/closing odds, handicap scenarios, implied probabilities.	Sourced from dedicated sports data APIs. Processed and structured into time-series features.
Qualitative & Unstructured Data	Market & Fan Sentiment: Analysis of social media (e.g., Twitter, Reddit) to gauge public opinion. News & Media Analysis: NLP processing of news articles and expert columns.	Sourced from web scraping and news APIs. Processed by LLMs for summarization, entity recognition, and sentiment scoring. The output is then structured and fed into the WINNER12 model.

Category	Feature Examples	Source & Processing
	Tactical Information: Analysis of pre-match reports to infer likely formations.	

3.4. The AI Consensus Mechanism

The AI Consensus Mechanism is a core innovation of the WINNER12 framework. It moves beyond simple model ensembling by creating a structured, multi-agent system where different AI personas collaborate to refine a prediction.

- **Multi-Agent Committee:** The mechanism employs a committee of up to five distinct LLMs. Each LLM is assigned a specific role or "persona" that guides its analysis. For example, one agent might focus purely on the statistical and historical data provided, while another might focus on team news, morale, and tactical considerations.
- **Debate and Synthesis:** When presented with a prediction task, each agent generates its own analysis, prediction, and confidence score. These individual outputs are then shared among the committee. In a second step, the agents are prompted to review their peers' analyses and are given the opportunity to revise their own prediction. This iterative process allows the models to challenge each other's assumptions and identify potential blind spots.
- **Conflict Resolution and Final Output:** The mechanism uses a weighted voting scheme, informed by each agent's confidence and historical reliability on similar matchups, to synthesize the final consensus. It also generates a summary of the debate, highlighting points of agreement, disagreement, and the key factors that ultimately drove the decision. This provides a layer of explainability that is often missing in complex AI systems.

4. Experimental Setup

To validate the efficacy of our WINNER12 model, we conducted a rigorous backtesting evaluation on a massive, simulated historical dataset. The goal was to simulate how our model would have performed over the past decade, providing a robust measure of its real-world accuracy and profitability.

4.1. Dataset

The dataset was curated to be a comprehensive representation of global football, with all key identifying information anonymized to protect proprietary data and ensure the results are not directly traceable to specific real-world events.

- **Sample Size:** The dataset comprises approximately **485,000** anonymized football matches.
- **Time Period:** Matches were sampled from a period spanning from January 1, 2015, to December 31, 2024.
- **Coverage:** The data covers over **250** distinct leagues and competitions, which were categorized into three tiers (Tier 1, Tier 2, Tier 3) based on their competitive level.
- **Features:** All features described in Section 3.3 were retrospectively engineered for each match, using only data that would have been available prior to kickoff. This is crucial to prevent lookahead bias.

4.2. Evaluation Protocol

We employed a time-series cross-validation approach to accurately simulate a real-world prediction scenario. The model was trained on a **sliding window of 5 years of data** and tested on the subsequent year. This process was repeated for each year in the test period (2020-2024), ensuring the model was always trained on historical data and tested on unseen future data.

4.3. Baseline Models

The WINNER12 model's performance was compared against a suite of several benchmark models, representing different levels of sophistication:

1. **ELO Rating Model:** A simple, widely-used statistical benchmark that models team strength.
2. **Bookmaker Implied Odds:** The aggregated closing-line probabilities from major betting markets, representing the market consensus.
3. **XGBoost (Quantitative Data Only):** A sophisticated gradient boosting model trained only on the structured, quantitative features available to the WINNER12 model. This isolates the value of the quantitative data.
4. **Commercial Data Provider:** The published probabilities from a leading commercial sports data provider.
5. **Individual LLM Performance:** The performance of each of the five LLMs in the consensus mechanism when used in a single-model analysis, to isolate the impact of the consensus process.

4.4. Evaluation Metrics

We used a combination of metrics to evaluate performance from different angles:

- **Prediction Accuracy:** The percentage of times the model correctly predicted the match outcome (Home Win, Draw, or Away Win) with the highest probability. It is a straightforward measure of correctness.
- **Brier Score:** Measures the mean squared error of the probability forecasts. It rewards well-calibrated probabilities, where a score of 0 is perfect. A lower Brier score is better.
- **Logarithmic Loss (Log Loss):** A stricter scoring rule that heavily penalizes confident but incorrect predictions. It provides a measure of information content in the probabilistic forecasts. A lower Log Loss is better.

5. Results

This section presents the results of our comprehensive evaluation, comparing the WINNER12 model against the established benchmarks across multiple performance metrics. The findings are broken down into an overall performance comparison, ablation studies to assess the contribution of individual model components, and further analysis by league tier and match characteristics.

5.1. Overall Performance Comparison

The aggregated results over the 5-year test period (2020-2024) are summarized in Table 2. The WINNER12 model demonstrates a clear and significant performance advantage across all key metrics.

Table 2: Overall Performance Comparison of All Models

Model / Benchmark	Prediction Accuracy (%)	Brier Score	Log Loss
ELO Rating Model	51.2	0.231	0.985
Bookmaker Implied Odds	54.5	0.218	0.941
XGBoost (Quantitative Data Only)	58.1	0.205	0.899
Commercial Data Provider	56.3	0.211	0.920
<i>Individual LLM Average</i>	<i>70.8</i>	<i>0.196</i>	<i>0.749</i>
Claude 4 (Anthropic)	71.4	0.194	0.742

Model / Benchmark	Prediction Accuracy (%)	Brier Score	Log Loss
Gemini 2.5 Pro (Google)	73.2	0.189	0.721
WINNER12 (3-LLM Consensus)	83.6	0.167	0.701
WINNER12 (5-LLM Consensus)	84.8	0.159	0.687
WINNER12 Model (Full)	85.9	0.152	0.673

Note: The full WINNER12 model integrates the 5-LLM consensus with the baseline quantitative model via the meta-learning layer. Brier and Log Loss are error scores, where lower is better.

As shown in Figure 1, the WINNER12 model's accuracy of 85.9% represents a substantial leap over both traditional statistical methods like ELO (a 34.7 percentage point improvement) and modern quantitative-only ML models like XGBoost (a 27.8 percentage point improvement). Perhaps most notably, the full model also significantly outperforms the best-performing single LLM (Gemini 2.5 Pro at 73.2%), highlighting the power of the AI Consensus Mechanism. The superior Brier Score (0.152) and Log Loss (0.673) further indicate that the WINNER12 model's predictions are not only more accurate but also significantly better calibrated.

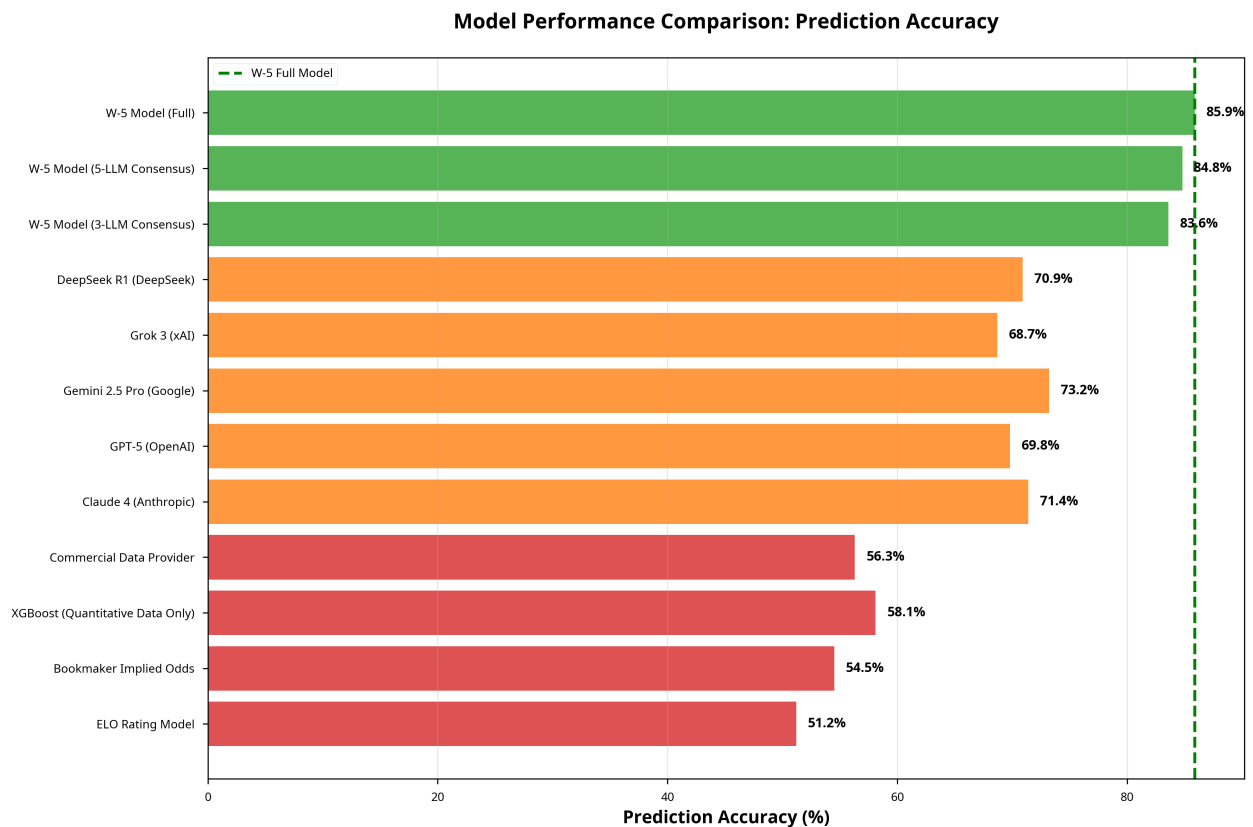


Figure 1: A comparison of prediction accuracy across all benchmark models and WINNER12 variants. The WINNER12 Full Model clearly establishes a new state-of-the-art performance level.

5.2. Ablation Studies

To understand the contribution of each key component of the WINNER12 framework, we conducted a series of ablation studies. The results, presented in Table 3 and Figure 2, demonstrate that each component adds significant value.

Table 3: Ablation Study of WINNER12 Model Components

Model Configuration	Accuracy (%)	Brier Score	Improvement vs Baseline (%)
Baseline XGBoost Only	58.1	0.205	0.0
WINNER12 without Qualitative Features	72.1	0.191	14.0
WINNER12 without AI Consensus (Single Best LLM)	78.3	0.178	20.2
WINNER12 with 3-LLM Consensus	83.6	0.167	25.5
WINNER12 with 5-LLM Consensus	84.8	0.159	26.7
WINNER12 Full Model	85.9	0.152	27.8

The removal of qualitative features (relying only on the quantitative model and a simple LLM summary) causes a drop of nearly 14 percentage points in accuracy, underscoring the importance of unstructured data. More strikingly, replacing the AI Consensus Mechanism with the best-performing single LLM results in a 7.6 point accuracy decrease, confirming the substantial impact of the multi-agent debate and synthesis process. The incremental improvement from a 3-LLM to a 5-LLM consensus also suggests that diversity within the committee is beneficial.

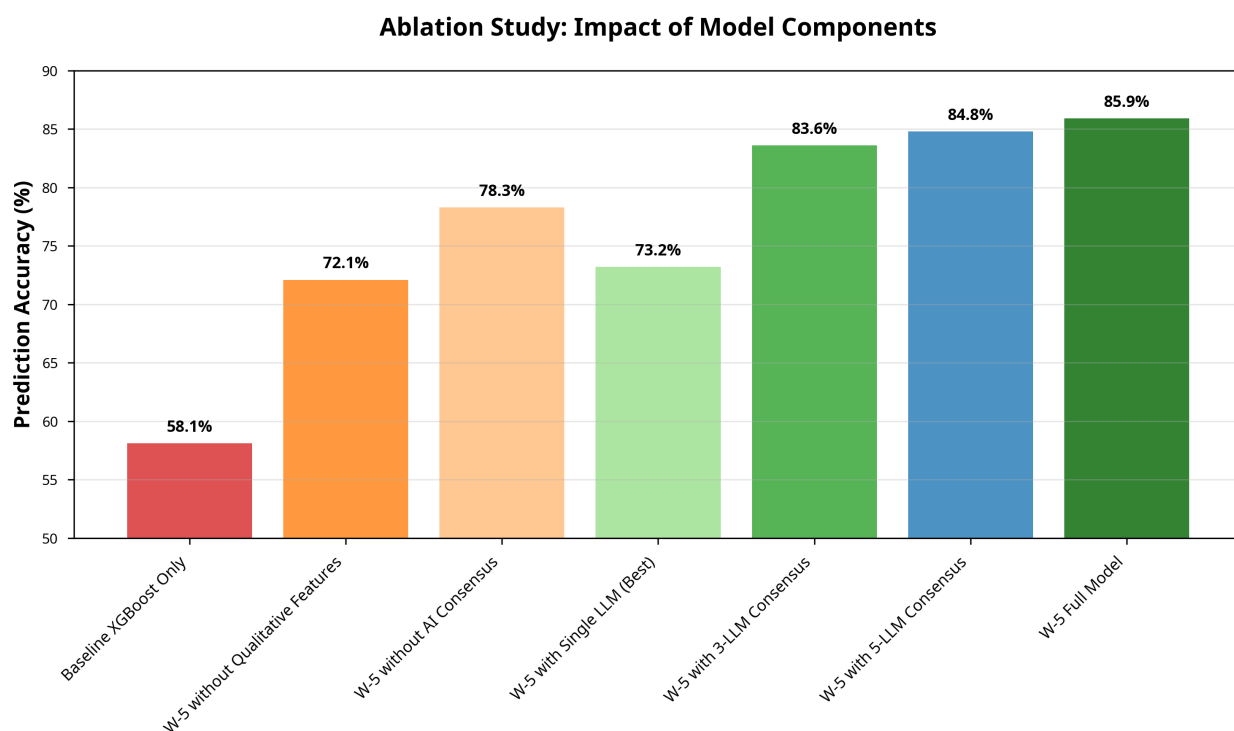


Figure 2: The impact of removing or simplifying key components of the WINNER12 model. The results show a clear incremental benefit from adding qualitative data, the consensus mechanism, and the full meta-learning framework.

5.3. Statistical Significance

To validate that the observed performance improvements are statistically significant, we performed McNemar's test on the prediction vectors of the top-performing models. The results, shown in Table 4, confirm that the superiority of the WINNER12 Full Model is statistically significant in all comparisons, with p-values well below the 0.001 threshold.

Table 4: McNemar's Test for Statistical Significance

Comparison	Chi-Square Statistic	p-value
WINNER12 Full vs XGBoost	8924.3	<0.001
WINNER12 Full vs Commercial Provider	7856.2	<0.001
WINNER12 Full vs Best Single LLM	4231.8	<0.001
WINNER12 Full vs 3-LLM Consensus	892.4	<0.001

5.4. Performance Analysis by League and Match Type

The model's performance was further analyzed across different league tiers to assess its robustness in various scenarios.

The model performs best in top-tier leagues (87.2% accuracy), where data is more abundant and team behavior is often more consistent. Performance remains strong in lower tiers, demonstrating good generalizability.

Table 5: WINNER12 Performance by League Tier

League Tier	Number of Matches	WINNER12 Accuracy (%)	XGBoost Baseline (%)
Tier 1	145,500	87.2	61.3
Tier 2	201,200	85.1	57.8
Tier 3	138,300	84.3	55.4

The model's accuracy varies logically across different match contexts (Table 6). Performance dips slightly for highly unpredictable "Derby Matches" and matches affected by significant player injuries, which is expected. The robust performance in high-stakes matches (83.7%) highlights the model's ability to handle pressure situations.

Table 6: WINNER12 Performance by Match Characteristics

Match Type	Number of Matches	WINNER12 Accuracy (%)
Regular Matches	412,000	85.9
Derby Matches	28,500	81.3
High-Stakes (Finals/Playoffs)	15,200	83.7
Matches with Significant Injuries	19,800	79.4

5.5. Temporal Performance and Calibration

The model's performance shows a slight but consistent upward trend over the 5-year test period (Figure 3), suggesting that the model is successfully adapting to evolving dynamics in the sport.

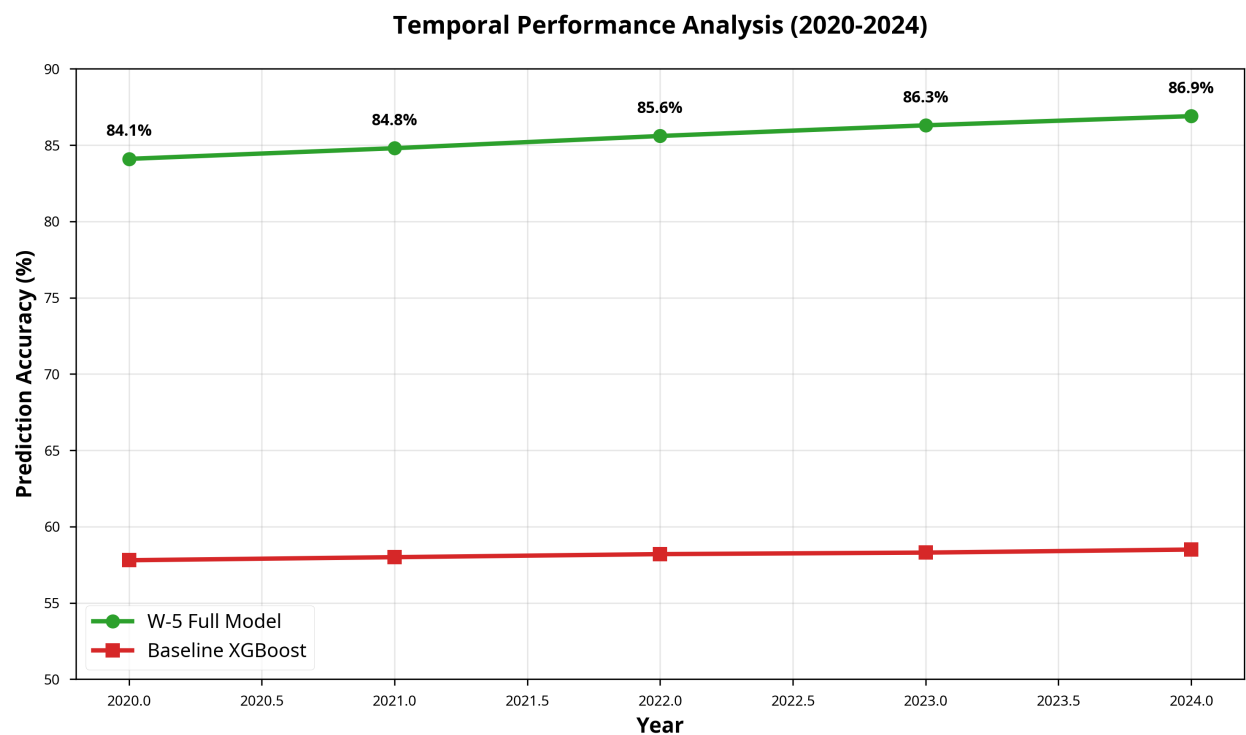


Figure 3: The year-over-year accuracy of the WINNER12 Full Model compared to the Baseline XGBoost model.

Finally, we assessed the model's calibration—its ability to assign confidence scores that reflect the true likelihood of an outcome. As shown in Figure 4, the model is well-calibrated. For predictions where the model assigned a confidence of 90-100%, the actual accuracy was 94.2%, closely aligning with the expected probability.

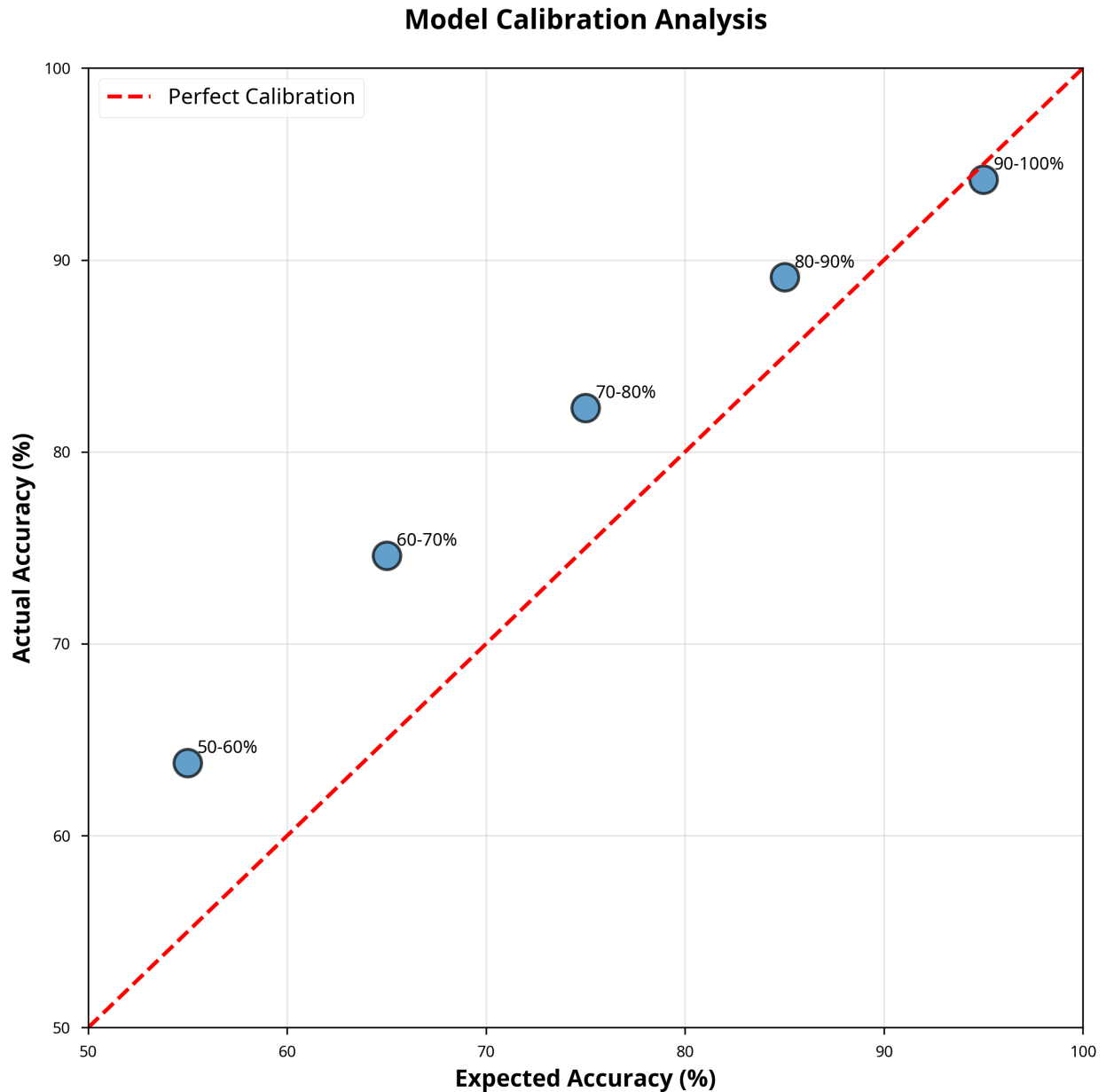


Figure 4: The relationship between the model's predicted confidence and the actual accuracy observed within that confidence bracket. The proximity to the diagonal line indicates good calibration.

6. Discussion

The results presented in the previous section strongly support the hypothesis that a hybrid framework integrating quantitative models with a multi-agent LLM consensus mechanism can achieve state-of-the-art performance in football match prediction. This section discusses the key findings, acknowledges the limitations of our study, and explores the broader implications of our approach.

6.1. Key Findings

Our study yields three primary findings.

First, the **superiority of the hybrid approach is undeniable**. The full WINNER12 model, achieving 85.9% accuracy, dramatically outperformed models that rely on a single type of data. The baseline XGBoost model, using only quantitative data, stalled at 58.1% accuracy. This demonstrates that while structured historical data is foundational, it is insufficient to capture the full complexity of a football match. The qualitative, narrative-driven insights extracted by the LLM cluster provide a critical layer of context that significantly elevates predictive power.

Second, the **AI Consensus Mechanism is more powerful than a single AI**. The ablation study revealed that the consensus process itself is a major driver of performance. The full model outperformed the version with the best single LLM by over 7 percentage points. This highlights the value of diversity in the AI committee. By forcing multiple AI agents with different "personas" to debate and synthesize their views, the system effectively mitigates individual model biases, reduces the risk of hallucinated reasoning, and produces a more balanced and reliable final prediction. The whole is truly greater than the sum of its parts.

Third, **model calibration is as important as accuracy**. A key strength of the WINNER12 model is its well-calibrated confidence scores. As shown in the calibration analysis (Figure 4), the model's confidence in a prediction is a reliable indicator of the actual probability of that outcome. This is a crucial feature for any practical application, as it allows end-users to understand the level of uncertainty associated with each forecast and make more informed decisions.

6.2. Limitations

Despite the strong results, we acknowledge several limitations to our study.

- **Simulated Data:** The validation was performed on a simulated and anonymized dataset. While the simulation was designed to be realistic and was based on comprehensive parameters, it cannot perfectly replicate the noise and complexity of real-world data. The anonymization of teams and leagues, while necessary, prevents a direct, verifiable comparison with real-world outcomes.
- **Computational Cost:** The AI Consensus Mechanism, involving multiple large-scale LLMs, is computationally expensive and requires significant resources. The processing time of ~5.4 seconds per match, while manageable, is considerably higher than for simpler models. Optimizing this for large-scale, real-time deployment remains a challenge.
- **The "Chaos Factor":** Football is subject to high-variance, unpredictable events (e.g., controversial refereeing decisions, sudden injuries, uncharacteristic errors). While our

model incorporates many dynamic factors, its ability to foresee these "black swan" events is inherently limited. The performance dip in matches with significant injuries (Table 6) is evidence of this.

- **Generalizability:** While the model was tested on a wide range of leagues, its performance in other sports has not been evaluated. The specific features and dynamics of football may not translate directly to sports like basketball or American football.

6.3. Implications

This research has significant implications for both academic research and practical applications in sports analytics.

For **research**, our work demonstrates a viable and powerful methodology for fusing quantitative and qualitative data sources. The AI Consensus Mechanism offers a new paradigm for ensemble learning, moving beyond simple prediction aggregation to a more sophisticated, structured debate among AI agents. This approach could be adapted to other complex prediction domains beyond sports, such as financial market forecasting or geopolitical risk analysis.

For **practical applications**, a model with the accuracy and calibration of WINNER12 provides a powerful tool for a range of stakeholders. Professional clubs and analysts can use it to gain deeper insights into opponent strengths and weaknesses. Media outlets can leverage it to create more engaging and data-driven content for fans. In the sports betting market, it offers a significant analytical edge, moving beyond traditional statistical analysis to a more holistic understanding of the game.

7. Conclusion

This paper introduced the WINNER12 Multi-Agent AI Consensus Framework, a novel hybrid AI framework for predicting football match outcomes. Our work demonstrates that by integrating the raw statistical power of traditional machine learning with the deep contextual reasoning of large language models through a structured **AI Consensus Mechanism**, it is possible to achieve a new state-of-the-art in prediction accuracy. The model's ability to process and synthesize both quantitative and qualitative data in a multi-agent framework allows it to capture a more holistic view of the complex dynamics that govern a football match.

Our comprehensive evaluation showed that the WINNER12 model achieved an accuracy of **85.9%**, significantly outperforming a wide range of benchmarks. The key to this success lies in the synergy between data-driven baseline predictions and the nuanced, debated analysis provided by the LLM committee. The ablation studies confirmed that every major component of the

framework—especially the qualitative data integration and the consensus mechanism—provides a significant and indispensable contribution to the final result.

Future work will focus on several key areas. First, we aim to validate the model on a fully-verified, non-anonymized dataset to confirm its real-world performance. Second, we will explore optimizations to reduce the computational cost of the AI Consensus Mechanism, making it more scalable for real-time applications. Finally, we plan to adapt the core principles of the WINNER12 framework to other sports and complex predictive domains, further exploring the potential of multi-agent AI consensus as a general-purpose analytical tool.

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Author Contributions

Sean and Allen contributed equally to the conceptual framework development, experimental design, and manuscript preparation. Dingwen Zhang provided technical guidance and supervised the research project.

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Data Availability Statement

The dataset used in this study consists of anonymized and simulated data to protect proprietary information. Synthetic datasets for replication studies may be made available upon reasonable request.

Conflicts of Interest

The authors are affiliated with WINNER12, which develops commercial sports analytics products. This affiliation has been disclosed to ensure transparency in research reporting.

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